

Graph Problems Related to Gate Matrix Layout and PLA Folding

Rolf H. Möhring*, Berlin

Abstract — Zusammenfassung

Graph Problems Related to Gate Matrix Layout and PLA Folding. This paper gives a survey on graph problems occurring in linear VLSI layout architectures such as gate matrix layout, folding of programmable logic arrays, and Weinberger arrays. These include a variety of mostly independently investigated graph problems such as augmentation of a given graph to an interval graph with small clique size, node search of graphs, matching problems with side constraints, and other. We discuss implications of graph theoretic results for the VLSI layout problems and survey new research directions. New results presented include NP-hardness of gate matrix layout on chordal graphs, efficient algorithms for trees, cographs, and certain chordal graphs, Lagrangean relaxation and approximation algorithms based on on-line interval graph augmentation.

Key words: linear VLSI layout architectures, gate matrix layout, PLA folding, interval graph augmentation, graph search, path width, vertex separation, complexity, approximation algorithms, matching with side constraints.

Graphentheoretische Probleme beim Gate Matrix Layout und PLA Folding. Der Artikel behandelt graphentheoretische Probleme, die bei linearen VLSI-Layout Architekturen wie Gate-Matrix-Layout, Programmierbaren Logischen Arrays und Weinberger Arrays auftreten. Zu diesen gehören u.a. die Einbettung von Graphen in Intervallgraphen mit kleiner Cliquengröße, Suchspiele auf Graphen, und Zuordnungsprobleme mit Nebenbedingungen. Wir diskutieren Folgerungen aus graphentheoretischen Ergebnissen für das VLSI Layout und geben eine Übersicht über neue Forschungsergebnisse. Hierzu gehören u.a.: die NP-Vollständigkeit des Gate-Matrix-Layout auf chordalen Graphen, effiziente Algorithmen für Bäume, Cographen sowie gewisse chordale Graphen, Langrange Relaxation und approximierende Algorithmen, die auf der On-line Konstruktion von Intervallgrapheinbettungen basieren.

1. Introduction

We study a class of graph-theoretic problems that arise in certain linear VLSI layout problems such as Weinberger arrays, gate matrix layout and PLA-folding.

The term *linear* refers to the fact that the most important degrees of freedom in the underlying VLSI architecture consist of linear (i.e. one-dimensional) arrangements of the relevant physical objects, the *gates*.

* Technical University of Berlin, Fachbereich Mathematik, Straße des 17. Juni 136, 1000 Berlin 12. This work was supported by the Deutsche Forschungsgemeinschaft (DFG).

In the most general form, an instance of such a linear layout problem consists of a $m \times n$ 0-1 matrix $M = (m_{ij})$ (the *net-gate matrix*), whose rows and columns represent the nets $N_1, \dots, N_m$ and the gates $G_1, \dots, G_n$ of the circuit, respectively.

The gates may be thought of as the basic electronic devices that are arranged linearly in a row, and the nets as realizing connections between them (details are given in Section 2). Net N_i must connect all gates G_j with $m_{ij} = 1$. Connections are realized rowwise by reserving for a given permutation of the gates (columns) for every net N_i the part of the row from the leftmost to the rightmost gate to which a connection must be established.

This can be expressed more formally by considering for a permutation π of the gates the *augmented net-gate matrix* $M_\pi = (\bar{m}_{ij})$ with

$$\bar{m}_{ij} := \begin{cases} 1 & \text{if there are Gates } G_r, G_s \text{ with} \\ & \pi(r) \leq j \leq \pi(s) \text{ and } m_{ir} = m_{is} = 1 \\ 0 & \text{otherwise} \end{cases}$$

Nets of the augmented net-gate matrix may share the same row (called *track*) if they have no gate in common. An assignment of augmented nets to tracks preserving this property is called a *feasible track assignment*.

The additional ones in M (with respect to the same column permutation of M) are called *fill-ins*. They are represented by a “*” to distinguish them from the given ones in M (see Figure 1.1 below).

The result of a permutation of the gates (*gate arrangement*) and an associated feasible track assignment is called a *layout*. Its area is proportional to $(\# \text{ gates}) \cdot (\# \text{ tracks}) = n \cdot (\# \text{ tracks})$.

So constructing an area-minimal layout (*optimal layout*) is equivalent to finding a gate arrangement and an associated feasible track assignment such that the number of tracks is minimum.

In matrix terminology, this leads to the following *matrix permutation problem* (MPP):

Given: A 0-1 matrix (net-gate matrix) M .

Problem: Find a permutation of the columns and an assignment of the augmented rows (nets) to tracks such that the number of tracks is minimum.

An example is given in Figure 1.1. We denote the minimum number of tracks by $t(M)$ and call a layout with $t(M)$ tracks an *optimal layout*. Due to the mentioned origin of this problem in VLSI-applications, there is an enormous body of papers on it. This article gives an overview on available results and new developments.

Section 2 deals with the VLSI background and models the linear VLSI layout technologies “Weinberger arrays”, “gate matrix layout” and “PLA folding” as instances of the general MPP. Some of these applications lead to restricted MPP’s in the sense that either the permutation of the gates (e.g. fixed first and last gate) or the assignment of nets to tracks (e.g. at most two per track) are restricted.